EOG Based Robotic Manipulator Control Using SVR Angle Prediction and SVM Direction Classification

Shrawan Rakesh Rane
Automation and Robotics
Mumbai , India
d2024.shrawan.rane@ves . ac.in

Dhanush Balasubramanian
Automation and Robotics
Mumbai , India
d2024.dhanush.balasubramanian@ves . ac.in

Rahulkumar Yadav
Automation and Robotics
Mumbai , India
2023.rahul.kumar@ves . ac.in

Vinod Santosh Kale
Automation and Robotics
Mumbai , India
d2024.vinod.kale@ves . ac.in

Kader Shaikh
Automation and Robotics
Mumbai, India
kader.shaikh@ves.ac.in

Sangeeta Prasanna Ram
Automation and Robotics
Mumbai , India
sangeeta.prasannaram@ves.ac.in

***Abstract*— The growing demand for hands-free human robot interaction offers an alternative to conventional robotic control methods. Electrooculography (EOG) provides a means of translating eye movements into control commands. This paper represents an EOG-Based robotic manipulator control system using support vector regression (SVR) for eye angle prediction and support vector machine (SVM) for direction classification. Horizontal and vertical EOG signals are acquired using the Bio Amp EXG pill and ADC interfaced with an ESP32-C3 mini. The processed signals are used to generate commands transmitted through UART to a shrike-FI ESP32 S3 controller, which operates a robotic manipulator. The developed system integrates EOG signal acquisition, real time serial data transfer and robotic manipulator through horizontal and vertical eye movements with joint selection and home position control available through the graphical interface. The experiment demonstrates validates the integration of bio-signal based interaction with physical robotic actuation with accuracy of SVR as 85% and SVM 70%.**

Keywords—Electrooculography (EOG), Robotic Manipulator, Support Vector Regression (SVR), Support Vector Machine(SVM), Eye Movement Detection, Machine Learning, ADS115, ESP32, UART Communication, Robot Feedback.

I. INTRODUCTION

Electrooculography (EOG) is a method that detects eye movements by measuring the electrical potentials produced when the eyes moves. These signals give data about horizontal and vertical eye motion and can be used as an alternative way to control machines. EOG-based control systems can be applied to technology, rehabilitation and hands- free operation of robots.

Traditional robotic manipulators are usually controlled with interfaces, like joysticks, switches or computer input devices. These interfaces may not work well for people who have limited physical mobility. An EOG-based control approach offers an option by turning eye movements into robotic commands allowing users to interact with robots without using hands.

This research introduces a robotic manipulator control system that combines bio-signal collection, machine learning and robotic movement. Horizontal and vertical EOG signals are captured with the Bio-Amp EXG pill. Converted into digital data by an ADS115 Analog-digital convertor. An ESP32-C3 Mini collects the signals. Sends the data to a computer.

II. RELATED WORK

Electrooculography (EOG) has been widely studied for hands- human-machine interaction, assistive technology and robotic control. Many studies have shown that eye movements and blinking patterns can be translated into control commands. However these studies vary in their signal-processing techniques, classification methods, calibration procedures, robotic platforms and control architectures.

Mogan [1] developed an EOG-based robot control system that uses eye blinking to send commands to a humanoid robot. Their method relies on real-time blink detection using amplitude thresholds and signal-processing techniques to reduce noise and handle baseline drift. The system identifies blinking patterns and uses a state-handling mechanism to manage robot actions. While this study proves that time EOG-based robotic control is possible it mainly focuses on generating commands through blinks rather than estimating eye angles continuously.

Abdel-Samei et al. [2] Created an EOG-based control system for a robotic arm using Arduino microcontrollers. Their setup includes an analog signal-conditioning circuit built around the AD620 instrumentation amplifier and the LM324 operational amplifier along with an NRF24L01 wireless transceiver. Horizontal and vertical eye movements are. Mapped to specific robotic commands. This research presents a hardware-focused approach to controlling a robotic arm through EOG. Still the control strategy depends on predefined movement categories of a learned continuous relationship between EOG signals and eye angles.

Perez Reynoso et al. [3] Introduced an EOG-based human-machine interface for real-time tracking of a robotic manipulator's trajectory. Their method models EOG signals using a multilayer neural network and uses genetic algorithms to fine-tune signal thresholds. A fuzzy inference system then maps the calibrated EOG values to coordinates for manipulator control. This work shows how important user-specific signal characteristics and calibration are in EOG-driven robotics. However their method differs from the combination of regression-based angle prediction and machine-learning-based direction classification.

Alshehhi et al. [4] Explored machine-learning-based robotic arm control using EOG signals. Their study compares Support

Vector Machine (SVM) Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) models for classifying eye movements. Using an EOG dataset and stratified five-fold cross-validation they report classification accuracies of 82.0% for SVM 90.3% for CNN and 93.2% for LSTM. These results highlight the potential of machine learning in recognizing eye movements.. The main focus is on discrete gesture classification not on combining continuous angle prediction with direction classification.

Zendehdel et al. [5] Proposed a hands- EOG-based control system for unmanned aerial vehicles using LSTM networks. The system detects gaze. Blink events. Users select commands via an interface and confirm them by blinking. The authors report a command-detection accuracy of 97.6% under conditions. This study shows how promising EOG-based interfaces can be for real-time control. However their application is limited to UAVs. Their method relies on sequential LSTM inference rather than combining SVR-based angle prediction with SVM-based direction classification.

A. Research Gap

The studies reviewed confirm that EOG-based robotic control is possible through blink detection, threshold-based directional commands, fuzzy inference and machine-learning-based gesture classification. Yet most of these approaches focus on command generation, personalized calibration or classifying eye-movement patterns.

The current work addresses this gap by combining eye-angle prediction with discrete movement-direction classification in one unified robotic control system. A structured visual target chart is used during data collection to link vertical EOG measurements with reference eye angles and movement labels. The signals are collected using the BioAmp EXG Pill, ADS1115 analog-to-digital converter and ESP32-C3 Mini.

The collected dataset is used to build two machine-learning models: Support Vector Regression (SVR), for predicting eye angles and Support Vector Machine (SVM) for classifying movement directions. Both model outputs are used to generate robotic control commands. The predicted eye angle is mapped to the corresponding robot angle and sent via an ESP32-S3-based Shrike-FI interface using UART communication and a 74HC126 buffer circuit to control the Hiwonder MaxArm robotic manipulator.

In addition the system includes robot-position feedback to assess movement accuracy. The robot angle derived from feedback is converted into an eye angle and compared with the SVR prediction to compute angular error. This provides a performance-monitoring mechanism to evaluate how predicted eye movements match actual robotic movements.

III. PROPOSED SYSTEM

The design is split into four parts: EOG signal Acquisition Signal processing, Machine learning prediction, Robotic manipulator control . The overall system architecture of our system are as follows.

A. Overall System Architecture

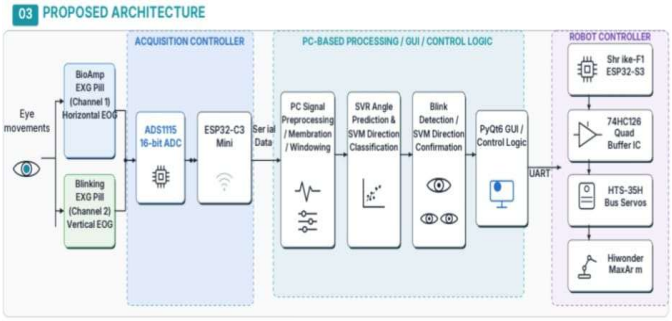


Fig. 1 The Overall architecture of the system

The proposed system uses the BioAmp EXG Pill to collect horizontal and vertical electrooculography signals that come from eye movements. The signals that are collected are analog. These analog signals are changed into digital using an ADS1115 analog to digital converter. The digital signals are connected to an ESP32-C3 Mini microcontroller. The digital information is sent to a computer. On the computer signal processing and machine learning analysis take place.

Two machine learning models are used to understand eye movements. The Support Vector Regression (SVR) model calculates the eye movement angle. The Support Vector Machine (SVM) model decides the direction of the movement. Both results are used to create the control instructions.

The predicted eye angle of the SVR model is mapped with the robot angles. The instructions that are created are sent to the Hiwonder MaxArm. This is done using an ESP32-S3-based Shrike-FI controller. The communication is done through UART. A 74HC126 buffer circuit is added to the servo communication interface. This helps with connecting the signals.

The robotic arm follows the instructions it receives. It sends back the joint position information. This information is changed into an eye angle. It is then compared with the eye angle that the SVR model predicted. The error is calculated, between the two angles is shown on a graphical user interface. This is used to watch how well the system is working and to check its performance.

B. EOG Signal Acquisition

The part of the system that deals with acquiring eye movement data collects the signals that happen when the eyes move up and down or left and right. A device called the BioAmp EXG Pill acts as the step in getting these signals. It connects to electrodes on the skin to get the EOG signals. Different channels are used to detect the vertical movements of the eyes.

The signals that are captured in an analog form are sent to a device called the ADS1115. This is a 16-bit converter that changes the EOG signals into digital numbers. The ADS1115 talks to a computer called the ESP32-C3 Mini using a method called I²C. This small computer takes the digital numbers from

both channels and sends the data to a bigger computer through a serial connection for more work.

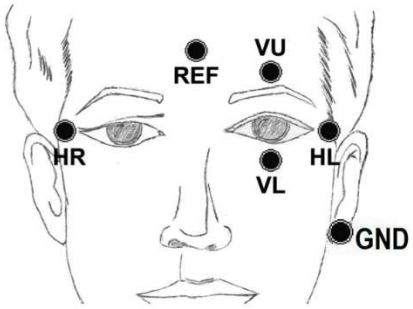


Fig 2. Electrodes placement for the EOG Collection

Electrode Placement for EOG signal acquisition for Horizontal and Vertical Signals

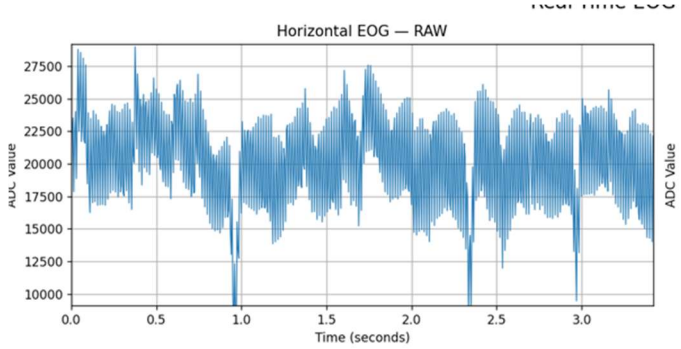


Fig.3 Raw Eog signal that we captured

Raw EOG signal is capture and filtering is done to like Noch filter to avoid 50Hz signal interference and the Band pass filter as EOG signal is in range of 0 – 30Hz so to pass that much signal only and reject the unwanted signal noise

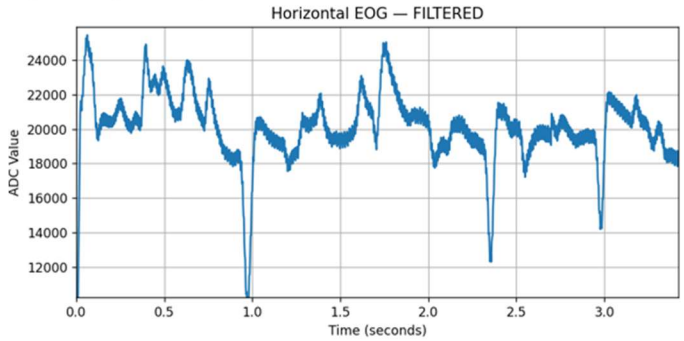


Fig.4 Filtered EOG Signal that we captured

First the process start with the signal acquisition so the EOG signal is in the range of the 0.05mv to 3.5mv to capture those we are using the BIO Amp exg pill which basically a amplifier it convert the low EOG signal into 2.5v to 5v as these data is send to the ADS115 which is the 16 Bit ADC we are using these to convert our continuous signal to digital signal the ADS is working at 250Hz sampling frequency as per Nyquist Criteria the sampling frequency must be twice or thrice the highest frequency so as per the suggested by bio medical signal processing we are keeping the sampling as 250Hz.then the espC3 mini reads data from the ADS via I2C protocol and send send the digital values to the system Via UART

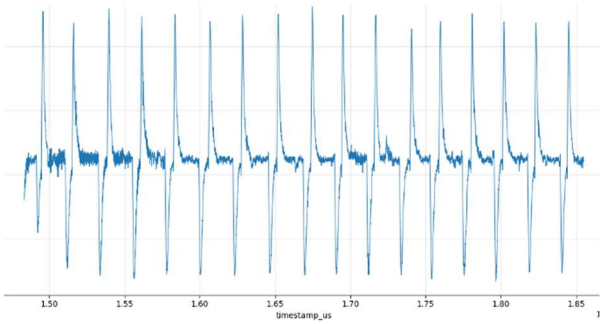


Fig.5 The EOG signal that we plotted from our collected dataset

Then the machine Learning part Comes into play here we basically the model basically has the has the window of 125 samples so it takes 125 samples at 250Hz at every 0.5sec as it is incremented by overlap of 50% so every time 62 new samples are used to make prediction and its update interval is 0.248 seconds for each window horizontal and vertical signal undergo feature extraction. From these the ML model Support vector Regressor (SVR) predicted the angle and the SVM give the direction the angles that is predicted by the SVR the predicted angle is converted into robot angle scaling is done and the data is send to the ESP 32 S3 Shrike FI board the shrike FI takes the movement command by the UART and give it to Buffer IC 74HC126 and the robot moves the scaled angles and feedback is taken for robot angle and the error of the SVR predicted angle and the Robot angles is calculated and shown on the UI Dashboard.

C. Eye Movemnet Preprocessing Using Machine learning

The EOG data that is collected is worked on by two models that use machine learning. These models are called Support Vector Regression and Support Vector Machine. Each of them has a role in understanding how the eyes are moving.

The Support Vector Regression model is used to find out the angle of eye movement from the EOG data. This angle is used as a guide to figure out where the robotic parts should move.

The Support Vector Machine model looks at the features of the signal. Decides which direction the eye is moving. It tells which movement category it's and gives the direction needed for the robot to follow.

The results from both models are used together to decide what the robot should do. This way the system uses both an angle reading and a direction choice.

D. Controlling the Robotic Arm

The results from the machine learning models are turned into commands for the robot. A system that maps angles is used. The angle that the Support Vector Regression model predicts is matched to the angle that the robot's part should move to. The Support Vector Machine result tells which direction the movement should be.

These commands are sent from the computer to a controller called the ESP32-S3-based Shrike-FI. This controller sends the commands to the Hiwonder MaxArm servos using a method called UART.

A part called the 74HC126 buffer is used in the connection to the servos to make sure the signals are correct. When the robot gets the commands it moves its parts to the places.

E. Measuring Position and checking errors

To see how well the robot is moving the system gets position data from the robot. This data comes back through the connection that the robot uses. It is shown on a screen.
The real angle of the robot's joint is changed back into an eye angle using the reverse of the angle map. This value is compared with the one that the Support Vector Regression model predicted to see how far off it is.
The error is found by subtracting the feedback angle from the Support Vector Regression angle. The formula is:

$$\boldsymbol{e_\theta = \theta_{SVR} - \theta_{feedback}} \quad \mathbf{(1)}$$

Where θ_{SVR} is the angle predicted by the Support Vector Regression model and $\theta_{feedback}$ is the angle from the robots actual position.
This error is shown on the screen along with the predicted and actual angles. It helps to see how accurate the robot is and how different the actual movement is from what was intended. This feedback is, for checking how well the system works, not for changing the commands that come next.

IV. Hardware Implementation

The proposed system consists of an EOG acquisition unit, a data acquisition microcontroller, a computer-based processing unit, and a robotic control interface. These components are integrated to acquire eye movement signals, process them using machine learning models, and execute the corresponding robotic movements.

A. EOG Acquisition PCB

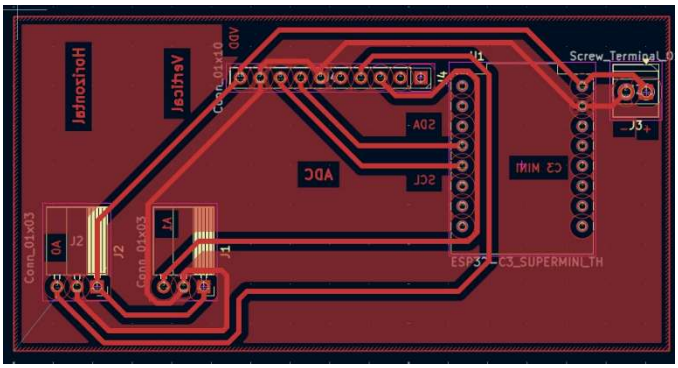


Fig 6. The EOG Acquisition PCB design

A custom PCB built around the ESP32-C3 Mini is created for EOG data acquisition. This board talks to the ADS1115 converter using I²C. The BioAmp EXG Pill supplies horizontal and vertical EOG signals and the ADS1115 turns them into digital values. The ESP32-C3 Mini reads these samples and

B. Robotic Control PCB

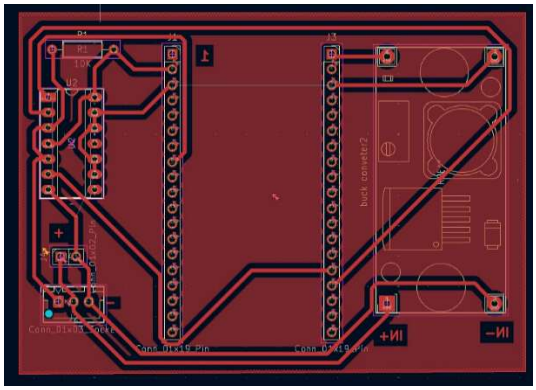


Fig 7. The Robot Control PCB design

A different custom PCB is built around the Shrike-FI controller for controlling the manipulator. This board gives the needed connections for talking between the computer and the robotic servo system. The Shrike-FI receives control orders from the computer over USB serial and talks to the HTS-35H bus servos with a half-duplex UART link. A 74HC126 buffer IC is added to help the servo signal connection

C. Computer- Based Processing Unit

The computer is the processing unit of the system. The computer takes EOG samples from the ESP32-C3 Mini. Runs signal processing and machine-learning inference in Python. The SVR model predicts eye-movement angles and the SVM model decides movement directions. Both results are used to create robotic control commands.

D. Hiwonder MaxArm Robotic Manipulator

The Hiwonder MaxArm is the robot used to show the proposed control method. Its HTS-35H bus servos move the joints based on commands from the Shrike-FI controller. The real joint-position feedback goes back, to the computer, where it

becomes an eye angle that is compared with the SVR prediction to find the error.

V. Digital Signal processing And Data Preparation system design parameters

After getting the EOG signals and converting them into data the horizontal and vertical samples are handled on the computer to get them ready for machine learning inference. This part is about reducing changes in the signal arranging the samples that were collected and making the right inputs for the SVR and SVM models.

A. Digital Signal Preprocessing

The EOG signals that are collected might have baseline drift, power-line interference and other types of noise. Digital filtering methods are used to make the signal better while keeping the parts about eye movement.

A 50 Hz notch filter is used to reduce power-line interference. A Butterworth bandpass filter with cutoff frequencies of 0.1 Hz and 10 Hz is used to keep the parts of the EOG signal.

B. Signal Windowing and Data Preparation

The EOG signals that have been filtered are put into sample windows that're good for machine learning. Using windows helps the system look at parts of the signal and give the right data to the models that have been trained.

The data that is ready is given to the SVR and SVM models. The SVR model uses it to predict eye angles. The SVM model uses it to figure out movement directions.

C. Machine Learning Input Generation

The EOG data that has been processed is changed to match what the machine learning models need. The SVR model gets the data to estimate the eye movement angles that're continuous. The SVM model uses the data to figure out the direction of movement.The results, from both models are then used to create robotic control commands.

D. Moving Average Smoothing

To further reduce short-term fluctuations and prediction jitter, a moving average filter with a window size of five samples is implemented. The filter calculates the average of consecutive samples to produce a smoother signal.

The moving average output is expressed as:

$$y[n] = \frac{1}{5}\sum_{k=0}^{4} x[n-k] \quad (2)$$

where (x[n]) represents the input sample and (y[n]) represents the filtered output.

Different window sizes were considered during implementation. A three-sample window provided insufficient smoothing, resulting in noticeable fluctuations, whereas a seven-sample window introduced additional delay. Therefore, a five-sample moving average window was selected as a compromise between signal stability and responsiveness.

In the moving average filter, the sample rate was set to five samples to ensure smooth robot movement. We tested the samples of 7 and 3; the robot moved smoothly with the first sample, but there was a system delay. With the sample of 3, the robot displayed more jitters. Through experimentation, we kept a sample of 5 for our moving average, which proves more effective for our system.

E. System design parametrs

The EOG acquisition system operates at a sampling frequency of 250 Hz, corresponding to a sampling interval of 4 ms. The ADS1115 ADC provides 16-bit resolution for digitizing the acquired analog signals. The acquired signals are processed using a 50 Hz notch filter to suppress power-line interference, followed by a fourth-order bandpass filter with a frequency range of 0.1–10 Hz. For machine learning, a processing window of 125 samples (0.5 seconds) is used with 50% overlap, resulting in a hop size of 62 samples between consecutive windows. These parameters are maintained to provide consistent signal acquisition and preprocessing for subsequent angle prediction and direction classification.

VI. Data Collection and Preparation

The EOG dataset was collected under the controlled experimental conditions using a structured visual target chart. A 5x5 grid was designed with the central point representing different eye movement angles at 5 degree intervals. The participant was seated approximately 45 cm away from the chart to maintain a consistent viewing distance throughout the recording process.

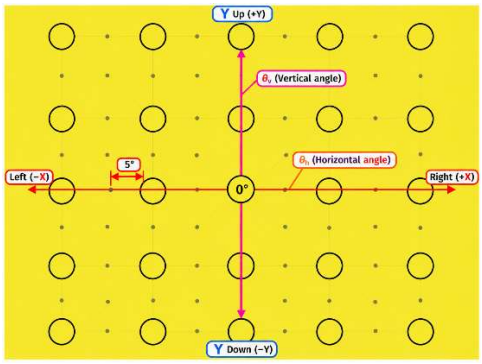


Fig 8. Dataset collection calibration chart

Horizontal and vertical eye movements were recorded by asking the participant to focus on specific target positions. Horizontal movements covered angles from -20 degree to +20 degree, including left and right directions, while vertical movements were recorded across the corresponding upward and downward target positions. Blink events were also recorded across the corresponding upward and downward target positions. Blink events were also recorded as a separate movement category.

At each target position, the EOG signals were acquired and converted into digital ADC measurements using the ADS115. The corresponding reference angle and movement direction were recorded to generate labelled samples for machine learning model training. A total of 23 datasets were collected with 1.8 million datapoints, covering left, right, up, down and blink movements. Each recording session one action lasted approximately 2 minutes 3 seconds

Our dataset therefore has columns like timestamp , horizontal and vertical ADC for the point at which user is looking at , target angle in which user is seeing and the label means the action (left, right, up, down) so it helps for getting information about each point in the grid

Reference angle caluculation

The reference angles for each target location on the visual chart are calculated based on the geometry of the target movement and viewing distance.

The horizontal reference angle is given by:

$$\boldsymbol{\theta_x = \tan^{-1}\left(\frac{x}{D}\right)} \; - \; (2)$$

The Vertical reference angle is given by:

$$\boldsymbol{\theta_x = \tan^{-1}\left(\frac{y}{D}\right)} \; - \; (3)$$

Angles have been converted from radians to degrees using the formula:

$$\boldsymbol{\theta_{degrees} = \theta_{radians} \times \frac{180}{\pi}} \; - \; (4)$$

Calculated angles are given as labels to the recorded ADC values. These labelled data sets make up the desired output for training the SVR regression model.

VII. SVR- Based Eye Angel predictions

The reason we have selected the SVR for continuous eye angle prediction because it can model the nonlinear relationships while maintaining low computation and less complexity during inference. Reason for not using the Neural network based regression approach is it require greater computation and long training time depending on the architecture we are following SVR provides comparatively lightweight solution for real time EOG based control system. Its suitability for small to medium sized datasets and its ability to generalize through the use of an appropriate kernel make it a practical choice for predicting eye angles from filtered EOG signals.

The Support Vector Regression (SVR) is used to predict continuous values of eye movement angles given the recorded EOG signals. SVR is a regressor which finds a function mapping from the input feature vector to the predicted continuous output.

This can be formulated as a function,

$$\boldsymbol{f(x) = w^{\top}\phi(x) + b} \; - \; (5)$$

Where $\boldsymbol{f(x)}$ represents the input EOG features and $\boldsymbol{\phi(x)}$ Denotes the feature space transformation **w** is the Regression coefficient vector, and (b) is the bias term

The SVR estimates a function that minimises the model complexity and the errors in the outputs beyond an (insensitive region. By using a kernel function, non-linear relationships between EOG features and desired angles can be captured.

The trained SVR model takes the extracted EOG features to give a predicted angle:

$$\boldsymbol{\hat{\theta} = f_{SVR}(x)} \; - \; (6)$$

If we are predicting both the horizontal and vertical angles separately, then we could train two individual regression models. This should be specified in the final implementation.

The predicted angle is then converted into an appropriate control signal for the robot, based on the robots allowable movement and implemented safety constraints.

Angle Prediction Evaluation
The angles predicted by the SVR can be evaluated by the comparing the actual angle and those for the target chart

Mean absolute error (MAE) is calculated as

$$\mathbf{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\theta_i - \hat{\theta}_i\right| \; - \; (7)$$

Root mean square error (RMSE) is calculated as

$$\mathbf{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\theta_i - \hat{\theta}_i\right)^2} \; - (8)$$

where N is the number of samples in the test data set, θ is the true angle, and $\hat{\theta}$ is the predicted angle.
The MAE tells us the average error we can expect from our model. The RMSE weights the larger errors more heavily. Both should be reported in degrees if we are regressing in degrees.

If predicting horizontal and vertical angles separately then these values should be calculated and reported separately.

VIII. SVM Based Direction Classification

The support vector machine (SVM) was selected for the movement direction classification. The combination of SVR and SVM enables the system to obtain both continuous eye-angles estimates and discrete movement- direction information for robotic manipulators control.

A Support Vector Machine classification is performed to determine the direction of the collected EOG data. This is achieved by performing a supervised classification problem using labelled examples within the dataset.

The classifier learns decision boundaries to separate between the different movement classes. These classes will depend on the dataset implemented and may consist of left, right, up and down movements.

The prediction can be given as

$$\hat{\boldsymbol{d}} = \boldsymbol{f}_{\mathbf{SVM}}(\mathbf{x}) \; - \; (9)$$

Where **f(x)** is the input EOG features and $\hat{\mathbf{d}}$ is the predicted movement class.

The SVM output gives a discrete directional command whilst the SVR gives a continuous angular estimate.

The classification output can be used to determine the desired direction to move in whilst the regression output determines what angle to move to. The exact command generation should reflect how you will implement the command, this may include determining which axis to move, what to do in a neutral state or implementing any movement constraints.

Direction Classification Evaluation
The performance of the SVM classifier can be evaluated using the labelled test data set. The classification accuracy can be found using,

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100$$

Accuracy can also be calculated for each direction class using precision, recall and F1 score. A confusion matrix can also be used to find any systematic errors. For example, movements to the left may have been classified as a right movement or vertical movements may have been confused.

IX. Overall System Working



Fig.9 Robot left movement

In the above Fig.9 robot is moving Left for the horizontal left EOG signal

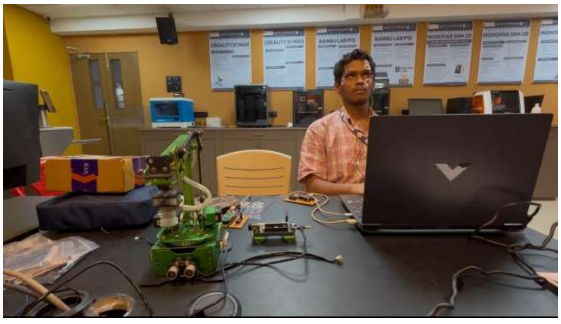


Fig.10 Robot Up movement

The above Fig.10 show robot moves the Up direction for the vertical Up EOG Signal



Fig.11 Robot Down movement

The above Fig.11 Robot Moves Down for Vertical Down EOG signal

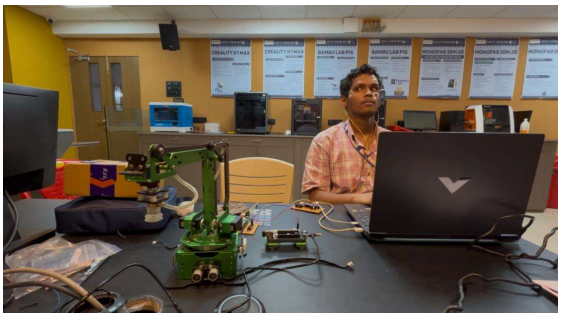


Fig.12 Robot Right movement

The Robot Moves Right for the Horizontal Right EOG signal



Fig.13 Robot Control GUI

The above is the GUI where we can see the robot feedback and the current position of the robot and the Error difference between the SVR predicted EOG angles and the feedback angles taken from the robot

Before pressing the start robot we need to perform calibration the calibration begins with 10 seconds of Centre-point fixation, followed by gaze recording at four points each in Left, Right, Up, down directions with 2 seconds per point. The user then looks at the Centre for another 5 seconds, followed by single-blink calibration for 10 seconds and double-blink calibration for 15 seconds. Then we start controlling the robot.

X. Results And Discusssion

A. SVR-Based Eye- Angle Prediction Results

The Support Vector Regressor (SVR) performance model is evaluated by comparing its predicted eye angle values with the corresponding actual reference values obtained from the prepared EOG dataset. The model uses Filtered EOG Signals to estimate Continuous Eye angle Variations, making it suitable for the regression based robotic control system

Table 1. SVR- Based Eye-Angle Prediction Performance.

Evaluation Metric	Obtained Values
R^2 Score	0.85
RMSE	38.78
MAE	30.90

We obtained R^2 score of 0.85 that our SVR model explains a part of the variation in the reference eye-angle values. The RMSE is 38.73 and the MAE is 30.90 it shows the prediction errors between the estimated eye angles and the actual eye angles. The results show that our SVR model can learn the relationship between filtered EOG signals and the corresponding eye-angle values.

Some deviations between the predicted eye angles and the actual eye angles may happen because of variations in EOG signal amplitude, electrode contact, baseline drift, electrical noise and differences in eye movements. These variations can change the consistency of the acquired signals. Therefore affect the regression output.

Overall the results show the potential of SVR for eye-angle estimation in the proposed EOG-based robotic control system. The obtained performance gives a basis for generating angle-dependent robotic manipulator commands. Yet further improvements, in calibration, signal processing and model optimization may help reduce the prediction errors.

B. SVM- Based Direction Classification Performance

We have evaluated the our SVM model to determine and classify eye movements into corresponding direction categories using Filtered EOG signals. The Classifier provide Movement direction information that is combined with the continuous eye angle prediction obtained from SVR model for robotic manipulator control

Table 2. SVM-Based Classification Performance

Evaluation Metric	Obtained Values
Accuracy	70%
F1 – Score	62%
Balanced Accuracy	65%

The SVM classifier got a classification accuracy of 70%. That means 70% of the samples that were checked were put into the group. The balanced accuracy is 65%. This shows how well the classifier works for each movement type. It also takes into account if some groups have samples than others. The Macro F1-Score is 0.62. This number tells us how good the classifier is for each group. It looks at both precision and recall.

There is a difference between the accuracy and the balanced accuracy. This means that the classifier might do better for some movement types than others. This could happen because

the EOG signals look similar, for some movements. It could also be because the signal strength changes or because the electrodes are placed in spots. Also some eye movements might have things in common.
The results show that the SVM classifier can tell which way the eyes are moving from the EOG signals. When it works with the SVR-based angle prediction model it gives both the direction and the exact angle. This is what the robotic manipulator control system needs. The classification results show that there is still room to get better. Improving the balance of the dataset making the signals clearer and working on the model more could help.

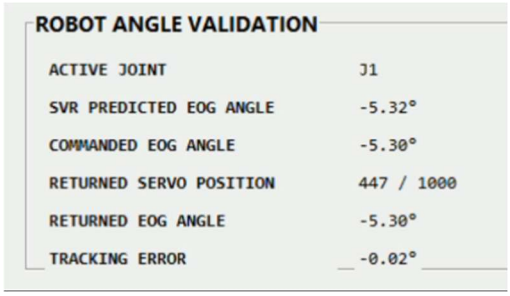


Fig 14. GUI Feedback and validation Section (Left Action)

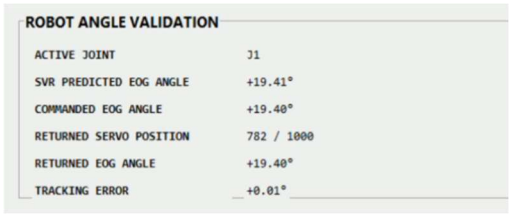


Fig 14. GUI Feedback and validation Section (Right Action)

The GUI Validation section where the feedback from robot is coming and the SVR predicted angle and and the robot angle is there from which the error is calculated.

XIII. Conclusion and Future Scope

A. Conclusion

This paper shows an Electrooculography (EOG)-based robotic manipulator control system that uses Support Vector Regression (SVR) to predict eye angles and Support Vector Machine (SVM) to classify eye-movement directions. The proposed system brings together EOG signal capture, digital signal processing, machine learning and robotic control into one framework.

We captured vertical EOG signals with the BioAmp EXG Pill and an ADS1115 ADC both connected to a custom ESP32-C3 acquisition PCB. The captured signals went to a PC for filtering and machine-learning processing. The SVR model reached an R^2 of 0.85. The SVM classifier scored 70% accuracy, 65% balanced accuracy and a Macro F1-score of 0.62.

We turned the predicted values into control commands for the Hiwonder MaxArm manipulator. The commands traveled through the Shrike-FI controller via UART and a 74HC126 buffer servo interface. The system also read real servo position feedback turned it into an eye-angle equivalent and measured the error between the prediction and the feedback.

The experiment shows that EOG-based eye-movement estimation can work together with machine learning to control a manipulator. The system lays a groundwork for hands-free human-robot interaction and assistive robotic uses.

B. *Future Scope*

The proposed system can be further improved in ways:

1. Improved Model Performance
We can fine-tune SVR and SVM models and test machine-learning methods to get better eye-angle predictions and movement-direction classifications.

2. Enhanced Dataset
Adding data more subjects, varied eye-movement patterns and different recording conditions will help the model generalise better and become more robust.

3. Real-Time Processing Optimization
Streamlining signal processing and model inference can cut response delay. Make the system faster.

4. Automatic Position Correction
We can use the servo feedback to create an automatic error-correction loop so that new robot commands adjust for the difference between predicted and real positions.

5. Multi-Joint Robotic Control
The system could grow to control robotic joints with horizontal and vertical EOG signals allowing more intricate manipulator motions.

6. Advanced Human-Robot Interaction
Adding gestures, like blink with high accuracy confirmation or command selection will make the system easier to use and more reliable.
Overall these upgrades can build an accurate, faster and dependable EOG-based robotic control platform, for assistive tech and hands-free robot use.

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